

JEE MAIN REVISION MASTER: MAGNETIC EFFECT OF CURRENT SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 209 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Magnetics - Core Application	209 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetics - Core Application (Very High Yield — 209 PYQs)

Core Principle: Key problem pattern in Magnetics.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ is placed in a magnetic field, $\vec{B} = B_0(\hat{i} + \hat{j})$." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "An insulated wire is wound so that it forms a flat coil with $N=200$ turns. The radius of the innermost turn is $r_1=3\text{cm}$, and of the out..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "The magnetic field at the centre of a current carrying circular loop of radius R is $16\mu\text{ T}$. The magnetic field at a distance $x=\sqrt{3}R$..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ is placed in a magnetic field, $\vec{B} = B_0(3\hat{i} + 2\hat{j})$. If $B_0 = 4 \times 10^{-3} \text{ T}$ and current $I = 100\sqrt{2} \text{ A}$, the torque experienced by the loop is ____ Wb.A. ($\pi = 3.14$)

- (A) $16 \times 10^{-5} \text{ k}$
- (B) $5024 \times 10^{-7} \text{ k}$
- (C) $5024 \times 10^{-7} \text{ i}$
- (D) $5024 \times 10^{-7} \text{ j}$ [Correct]

Correct Answer: D

Solution Analysis: When a current-carrying loop is placed in an external magnetic field, it experiences a torque ($\vec{\tau}$). The magnetic properties of the loop are represented by its magnetic dipole moment ($\vec{m}$), defined as : $\vec{m} = I\vec{A}$ Where I is the current and $\vec{A}$ is the area vector of the loop. The direction of the area vector is perpendicular to the plane of the loop, given by the unit normal $\hat{n}$. $\vec{A} = A\hat{n}$. The torque is given by the cross product of the magnetic moment and the magnetic field : $\vec{\tau} = \vec{m} \times \vec{B} \Rightarrow |\vec{\tau}| = IA\sin\theta$